2020-2021 FALL SEMESTER HYDRAULICS COURSE 4rd SHORT EXAM Date:02.01.2021

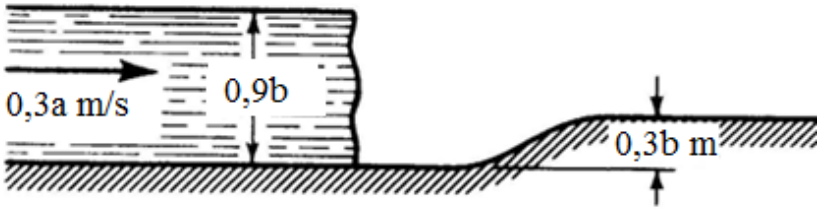
Name: **Surname:** **Student ID:**

Underline the last two digits of your student number you wrote above. The last two digits of your student ID number will be the "**ab**" value.

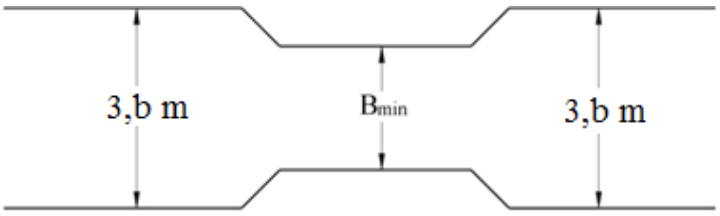
QUESTION (25P)

In the rectangular channel shown in the figure below the water depth is 0.9 m and moving at a speed of 0.3 m / s. The channel width is 3,b m.

a) If a hump (step) of 0.30 m is placed at the bottom of the channel. What will be the water depth above the hump? Draw the water surface profile.



b) For the channel contraction only (without any hump on the channel bottom), determine the possible minimum width of the channel that allows the flow passing the contracted section without changing the regime for the same upstream flow conditions as above? Draw the water surface profile.



c) In addition to the contraction (in (b)), if a hump with a height of 0.3 m is placed on the bed, determine the water depths before the contraction and at the contracted region. Draw the water surface profile.